CustomerSupport — Identity-Sensitive Model Examples

1 Case Introduction

This example is drawn from a large-scale deployment of a generative AI assistant at a Fortune 500 software company's customer support operation, studied by Brynjolfsson et al. [2023]. The company employed over 5,000 customer support agents handling technical issues via live chat. A GPT-based conversational AI assistant was introduced through a staggered rollout, providing agents with real-time suggestions during customer interactions—recommended responses, diagnostic steps, and resolution strategies. Agents could accept, modify, or ignore these suggestions at their discretion.

The deployment produced striking asymmetric outcomes: overall productivity increased by 14% (measured as issues resolved per hour), but this average masked a fundamental divergence. Novice agents improved by approximately 34%, while high-skill agents experienced minimal or slightly negative effects. The authors provide suggestive evidence that the AI "disseminates the best practices of more able workers and helps newer workers move down the experience curve"—in effect, the system captured expert behavioral patterns from historical data and redistributed them to less experienced agents.

In our models, Jake represents a veteran agent whose professional identity is grounded in rapid pattern recognition, high first-contact resolution rates, mentoring authority over junior colleagues, and a distinctive communication style developed through years of practice. Mia represents a novice agent whose baseline identity is still forming—she depends on scripts and escalation protocols, and her professional self-concept is aspirational rather than established. Support Operations is the organizational position responsible for throughput metrics, quality assurance, and workforce management.

Unlike the CleverLoan and NeuroYou cases, which are drawn from ethnographic or qualitative research with named individuals, the Brynjolfsson study is a large-scale quantitative analysis with anonymous agents. Jake and Mia are constructed as representative actors whose identity profiles are grounded in the study's empirical findings—specifically, its disaggregated productivity data, communication convergence patterns, and durable learning effects. The narration is transparent about this interpretive step.

2 Identity Tensions

The AI assistant generated identity effects that were invisible to the organization's productivity-oriented evaluation but analytically significant from an identity perspective:

- **For veteran agents (Jake):** The AI extracted Jake's accumulated expertise—his diagnostic patterns, resolution strategies, communication approaches—and redistributed them to the entire workforce without his participation, recognition, or consent. His skill differential, the foundation of his professional distinction, was systematically compressed. His informal mentoring role, through which he shaped junior agents' development, was displaced by the AI's automated dissemination of best practices. The organization measured a productivity gain; Jake experienced an erosion of the expertise-based identity that gave his work meaning.

- **For novice agents (Mia):** The same system accelerated her competence development dramatically. She gained access to expert-level resolution strategies immediately upon deployment, and—critically—retained measurable skill improvements even during periods when the AI was unavailable (durable learning). However, this rapid development was mediated entirely through AI dependency rather than through mentoring relationships with experienced colleagues. Her emerging professional identity was shaped by the AI rather than by the professional community.

- **For the customer relationship:** Customer sentiment improved and escalation rates decreased. But the communication patterns driving this improvement converged across all agents—the AI homogenized interaction styles by steering everyone toward the same optimal patterns. The distinctive relational qualities that individual agents brought to customer interactions were flattened.

This case is analytically distinctive because the identity effects were *unintended side effects* of a productivity-optimized deployment. Nobody designed for identity erosion; nobody mitigated it; the organization's evaluation framework—issues resolved per hour—was structurally incapable of surfacing it. The case thus demonstrates a different trajectory from the other three: **unintended homogenization**, where functional success coexists with unaddressed identity consequences.

3 'As-Is' SD Model

What this model conveys:

- **Agents, roles, and positions:** *Jake* and *Mia* are person-based agents who instantiate (INS) generic agents—*Veteran Support Agent* and *Novice Support Agent*, respectively. The *AI Chat Assistant* is a system agent. *Customer* is an external agent. The *Support Operations* position COVERS one formally recognized role:

  - **Supporting Customer**—the core operational role: receive chat, diagnose issue, resolve, close. Both the Veteran and Novice Support Agents PLAY this role. This is the only role the organization formally measures (issues resolved per hour), and where the AI's augmentative effect is evaluated.

  Two additional roles appear in the model as *informal* role boundaries—they are **not** COVERS'd by Support Operations:

  - **Escalation Handling**—managing complex or escalated cases that exceed routine resolution capability (multi-system failures, high-frustration customers, edge-case diagnostics). Only the Veteran Support Agent (Jake) PLAYS this role; Mia does *not*. The Brynjolfsson findings document that the AI *reduces escalation rates*—improving customer sentiment and enabling novice agents to resolve cases that would previously have required escalation. This means the AI is compressing the volume of work that flows through Escalation Handling, silently eroding a role that was structurally distinctive to experienced agents like Jake.
  - **Agent Mentoring**—guiding and socializing junior agents: modeling diagnostic reasoning, demonstrating communication style, explaining edge cases, providing feedback on handled chats. Jake's PLAYS link to this role remains structurally intact (shown as a solid line), but the role's function has been *functionally displaced*: Mia now receives expert-derived guidance from the AI Chat Assistant rather than from Jake. The displacement is invisible precisely because the structural link persists while the functional dependency has rerouted.

The informal status of Escalation Handling and Agent Mentoring is analytically significant: because neither role is formally COVERS'd by Support Operations, the organization has no metrics, no recognition mechanism, and no governance structure for these roles. The AI can erode them without triggering any organizational response—the only role the organization monitors is Supporting Customer, and that one is augmented, not eroded.

- **Role portfolio asymmetry—the structural key:** With three roles visible in the model, Jake and Mia are no longer structurally equivalent. Jake's professional identity is constituted across three roles (Supporting Customer + Escalation Handling + Agent Mentoring); Mia's involves only one (Supporting Customer). The AI erodes *two of Jake's three roles*—Escalation Handling through reduced escalation volume, Agent Mentoring through complete displacement—while augmenting both agents within the shared role. This asymmetry is the structural explanation for the Brynjolfsson study's central finding: why the same AI tool produces disproportionately positive effects for novices and minimal-to-negative effects for experts. Modeling only one role would make this invisible.

- **Core dependencies:** *Customer* depends on *Jake/Mia* for **Customer Issues be Resolved** (goal). Critically, the softgoal direction is *reversed*: *Jake* and *Mia* each depend on *Customer* for **Customer Trust be Sustained** (softgoal). This is an identity dependency, not a functional one—agents do not need customer trust to resolve tickets, but they need it to maintain their professional self-concept. Jake's version is grounded in years of earned expertise; customer trust validates his standing as a skilled practitioner. Mia's version is aspirational; customer trust signals that she is becoming competent. The AI's communication homogenization undermines both: when all agents sound alike, customers cannot distinguish Jake's expertise from Mia's AI-mediated responses, and Mia cannot be certain whether the trust she receives is directed at her or at the AI-generated communication she delivers. This reversed direction makes visible why communication convergence is an identity problem, not just a stylistic one.

  The model shows differentiated resource dependencies on the AI: *Mia* depends on *AI Chat Assistant* for **Communication Templates** (resource)—the AI provides phrasing, tone, and diagnostic scripts that shape her customer interactions. *Jake* depends on *AI Chat Assistant* for **Resolution Suggestions** (resource)—diagnostic and resolution recommendations that he can accept or override based on his judgment. This differentiation makes the homogenization mechanism precise: it operates primarily through the Communication Templates that shape Mia's professional voice, while Jake's relationship with the AI is mediated through suggestions he retains discretion over. *Support Operations* depends on both agents for **Throughput Targets be Met** (goal) and **Customer Satisfaction be Maintained** (softgoal). Within the informal *Escalation Handling* role boundary, *Support Operations* depends on *Jake* for **Complex Cases be Resolved** (goal); *Customer* depends on *Jake* for **Difficult Customer Issues be Resolved** (goal); and *Mia* depends on *Jake* for **Escalation Assistance** (resource) when she hands off cases she cannot resolve alone. The AI reduces how often these dependencies are activated, compressing Jake's distinctive contribution. Within the informal *Agent Mentoring* role boundary, *Mia* previously depended on *Jake* for **Guidance on Complex Cases** (resource) and **Professional Socialization be Sustained** (softgoal). This dependency has been functionally *rerouted* to the AI Chat Assistant—Mia's dependency on Communication Templates now serves the function that Jake's mentoring once served.

- **The invisible extraction:** The critical analytical feature of this model is the dependency that is *not* represented but structurally present: the AI Chat Assistant's effectiveness depends on historical behavioral data from high-skill agents like Jake. Expert knowledge flows *into* the AI (as training data) and *out* to novice agents (as suggestions), but this extraction has no corresponding dependency in the model. Jake is not a recognized *dependee* for the knowledge that powers the system. This one-directional flow—extraction without reciprocation—is struc-

turally invisible in a conventional SD model, yet it is the mechanism through which expert identity is eroded.

- **Displaced mentoring as functional displacement:** The model shows Jake's PLAYS link to Agent Mentoring as a solid line—the role has not been formally reassigned. Yet upon analysis, the role's function has been displaced: Mia no longer seeks Jake's guidance because the AI provides it through Communication Templates. The displacement is invisible precisely because the structural link persists while the functional dependency has rerouted. This is analytically distinctive: unlike CleverLoan, where the destructive behavior was visible, or NeuroYou, where professional conflict surfaced tensions, here the mentoring displacement generates no organizational signal at all.

- **Homogenized communication through differentiated pathways:** The model shows that homogenization operates through different resource dependencies for each agent. Mia depends on the AI for **Communication Templates** that directly shape her professional voice—this is the primary homogenization channel. Jake depends on the AI for **Resolution Suggestions** that he retains discretion over, but the convergence documented in the Brynjolfsson study suggests that even partial adoption steers communication patterns toward uniformity. The differentiated resource profiles make the mechanism more precise than a blanket "everyone gets the same templates" reading: the AI shapes Mia's voice directly and Jake's voice indirectly, with the net effect being convergence across both agents' communication styles.

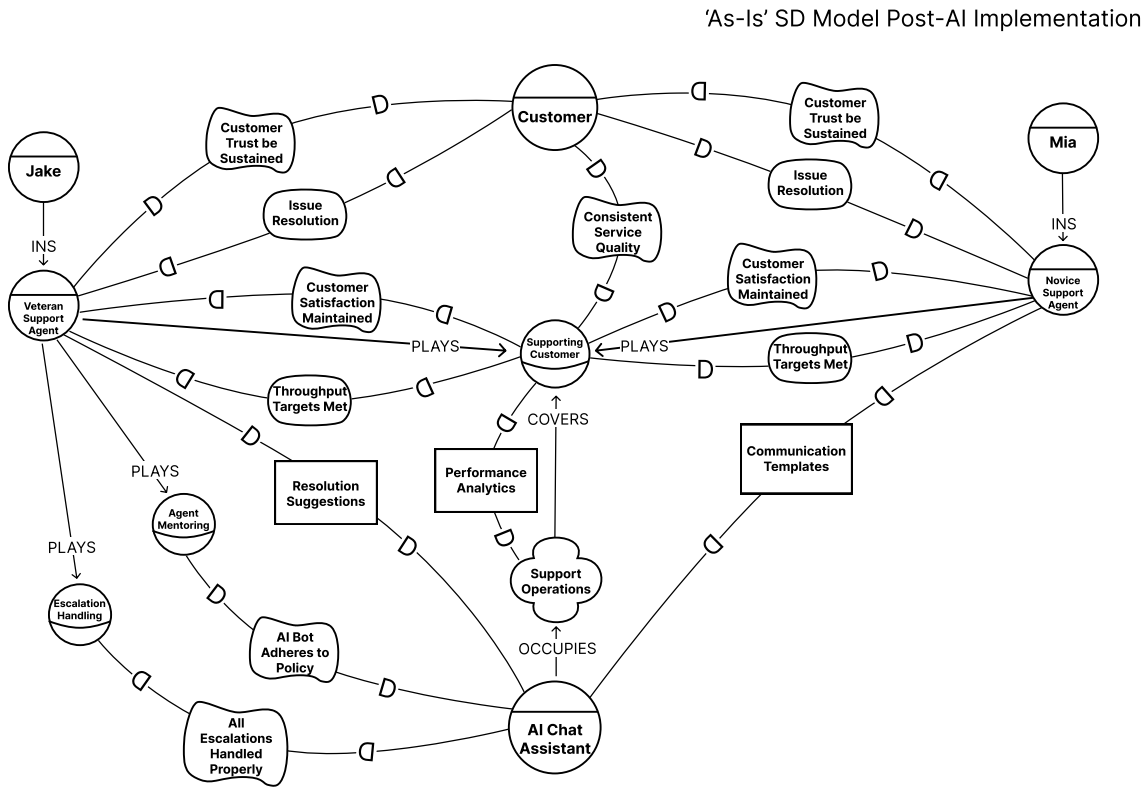

Figure 1: As-Is SD model—customer support with AI chat assistant.

Key analytical point. The As-Is SD reveals a system that appears functionally healthy—all goal dependencies are satisfied, throughput targets are met, customer satisfaction is maintained—yet harbors four identity-level pathologies invisible to conventional evaluation: (1) asymmetric knowledge extraction without recognition, (2) functional displacement of Agent Mentoring from

Jake to the AI (while Jake's structural PLAYS link remains intact, making the displacement invisible), (3) compression of Escalation Handling volume that erodes expert-distinctive work, and (4) communication homogenization through differentiated AI resource dependencies. The three-role structure makes the asymmetric impact *structurally visible*: Jake's identity spans three roles and the AI erodes two of them; Mia's identity involves one role and the AI augments it. A single-role model would hide this entirely. The formal/informal role distinction deepens the diagnosis: the only role the organization formally recognizes and measures (Supporting Customer) is the one the AI augments; the two roles that constitute Jake's distinctive identity (Escalation Handling, Agent Mentoring) are informal and thus invisible to organizational metrics. The SD model's value here is diagnostic: it makes the *structure* of these concerns visible through role and dependency analysis, even though the organization's performance metrics register only success.

What this reveals beyond CleverLoan and NeuroYou. The CleverLoan As-Is model (LC-2) showed divergent identities within the same role facing a *substitutive* AI. The NeuroYou As-Is model (NY-2) showed bilateral identity threats across *different professions* collaborating on a shared system. The CustomerSupport As-Is SD shows a different pattern: an *augmentative* AI that produces asymmetric identity effects not through task substitution but through **role portfolio erosion and knowledge redistribution**. The formal/informal role distinction adds a structural explanation for why this erosion is invisible: the organization's COVERS relationship extends only to Supporting Customer—the role the AI augments—while the two identity-constitutive roles (Escalation Handling, Agent Mentoring) remain informal, uncovered, and unmeasured. The identity threat is not "your task is being automated" (CleverLoan) or "your professional paradigm is being challenged" (NeuroYou), but "your distinctive roles are shrinking while your routine role is augmented, and nobody notices because the metrics only measure the formal role."

4 Identity-Sensitive 'To-Be' SR Model

What this model conveys:

This Strategic Rationale model presents an identity-sensitive redesign of the AI-assisted customer support system. It shows the internal reasoning of each actor in a reconfigured deployment that addresses the identity concerns surfaced in CS-1—not by removing the AI, but by restructuring how agents interact with it and how the organization recognizes the identity dimensions of AI-mediated work. Where CS-1 and CS-2 describe the same obligation, **dependum wording matches** (e.g., *Customer Issues be Resolved*, *Throughput Targets be Met*). CS-1 supplies **dependencies** and interpretive notes; CS-2 adds **means-ends**, task decomposition, and **contribution links** so identity trade-offs among alternatives are **explicit in the model**. Full qualitative label propagation (Chung et al., 2000; Horkoff, 2006) is **deferred**; the CS-2 specification documents structure only for this revision.

Construct integration (CS-1 ↔ CS-2)

Construct	**CS-1 (SD)**	**CS-2 (typal + SR)**
Core customer goal	D1, D2: *Customer Issues be Resolved*	**D1**; **J-G1**, **M-G1**
Identity softgoal (trust)	D3, D4: *Customer Trust be Sustained*	**D2**; **J-S1**, **M-S3**
Throughput / org expectations	D8, D9: *Throughput Targets be Met*	**D7**; **J-G1**, **M-G1**, **J-T4**
Org → AI	D10: *Performance Analytics*	**D9**; AI tasks
Delegation (ISA)	(implicit)	**D3**, **D4**
Human ↔ AI augmentation	D6–7 CS-1	**D5**, **D6**; **J-T4**, **AI-T1–T3**, **M-T1–T4**
Role formalization	R2/R3 informal	**D10–D15**; R1–R3 SR
Relational closure	(implicit)	**D16–D18**

Yu (1995) typing applies throughout CS-2: every *Goal* is an achievement state (satisfied / not satisfied for the stated scope); *Softgoals* carry qualities and partial satisfaction (see CS-2 spec: Goal vs. softgoal discipline). Hard goals **J-G1** and **M-G1** (*Customer Issues be Resolved*) align with first-line closure toward **R1-G1** (*Chat Service Targets be Met*) and typal **D1** / **D11**; **J-T2** with R2 tasks realizes **R2-G1** (*Expert Case Targets be Met*, **D13**); **R3-G1** (*Staff Development Targets be Met*) under Alternative B uses **J-T3**, **R3-T1**/**R3-T2**, **M-T4**, **D14**/**D15**. **AI-G1** (*Support Augmentation Targets be Met*) is realized by **AI-T0** (**D4**). Primary realizations (Alternative B): R1—**J-T1**, **M-T1**–**M-T4**, **AI-T1**/**AI-T2**; R2—**J-T2**, **R2-T1**/**R2-T2**; R3—**J-T3** plus **R3-T1**/**R3-T2**, Mia **M-T4**.

Role structure in the redesign

The three roles identified in CS-1 are preserved, but their organizational status and relationship to agents are fundamentally restructured. The redesign introduces a three-level **agent type hierarchy**: *Customer Support Agent* is the abstract agent type that OCCUPIES Support Operations. Two specializations exist: *Human Support Agent* (ISA Customer Support Agent) and *AI Support Agent* (ISA Customer Support Agent). Concrete agents are instances: Jake and Mia INS Human Support Agent; AI Chat Assistant INS AI Support Agent. Through ISA and INS inheritance, every concrete agent—human and AI—inherits the OCCUPIES relationship to Support Operations. All role access flows through the chain: INS → ISA → OCCUPIES → COVERS. No direct PLAYS links exist in the redesign.

This replaces the CS-1 structure, where agents reached roles through direct PLAYS links (via INS intermediaries), and only Supporting Customer was COVERS'd by Support Operations. The redesign extends COVERS to all three roles:

- **Supporting Customer** remains the shared operational role, still augmented by the AI—productivity gains are preserved. All agents access it through the inheritance chain to Support Operations.
- **Escalation Handling** is **formalized** under Support Operations' COVERS scope. In CS-1, the AI was compressing this informal role's volume with no organizational signal. Now that the organization COVERS it, escalation activity is a formal metric—volume reduction becomes a visible signal, not a silent erosion.
- **Agent Mentoring** is **formalized and restored** under Support Operations' COVERS scope. In CS-1, Jake's direct PLAYS link remained structurally intact but was functionally displaced—Mia received guidance from the AI rather than from Jake. The redesign formalizes the role so that mentoring activity is organizationally measured and recognized, and restores its function through Jake's AI-augmented mentoring tasks.

The elimination of direct PLAYS links is the structural mechanism that prevents the identity erosion diagnosed in CS-1. When agents access roles through the INS → ISA → OCCUPIES → COVERS chain, the organization mediates that access: it can measure performance across all three roles, recognize contributions within each, and detect erosion if a role's function diminishes. The ISA split (Human vs. AI Support Agent) places both human and AI agents under the same organizational governance while preserving the type distinction that grounds identity analysis: human support agents carry person-based identity; AI support agents do not.

Role boundaries R1–R3 (internal elements)

R1 Supporting Customer. R1-G1 *Chat Service Targets be Met* is a *hard goal* in the sense of Yu (1995): an achievement state that either obtains or does not (operational targets met or not met; aligned with D11), not a matter of degree on the goal node—qualities such as perceived service excellence are carried by softgoal R1-S1. Task R1-T0 *Execute Supporting Customer Workflow* (AND-decomposes R1-T1/R1-T2, means-ends to R1-G1): R1-T1 *Receive and Diagnose Issue*, R1-T2 *Resolve and Document*; R1-S1 *Service Quality be Maintained.*

R2 Escalation Handling. R2-G1 *Expert Case Targets be Met* (hard goal; **D13**); tasks R2-T1 *Receive and Diagnose Complex Case*, R2-T2 *Resolve and Document Complex Resolution*; softgoal R2-S1 *Expert Resolution Quality be Maintained.*

R3 Agent Mentoring. Means-ends structure for R3-G1 and tasks under Alternative B are detailed below (R3-T1, R3-T2; R3-S1, R3-S2).

Strategic dependencies (typal registry D1–D18)

Concrete agents instantiate typal links in SR; arrows use HSA/AISA/CSA. Abbreviated registry (full spelling in `CS-2_ToBe_SR_specification.md`):

	Depender → Dependee	**Dependum** (type)
D1	Customer → CSA	*Customer Issues be Resolved* (goal)
D2	HSA → Customer	*Customer Trust be Sustained* (softgoal)
D3	CSA → HSA	*Human Service Mandate be Fulfilled* (goal)
D4	CSA → AISA	*AI Augmentation Mandate be Fulfilled* (goal)
D5	AISA → HSA	*Expert-grounded Inputs* (resource)
D6	HSA → AISA	*Transparent Augmentation* (resource)
D7	P1 → HSA	*Human Workforce Commitments be Met* (goal)
D8	HSA → P1	*Identity-fair Conditions* (softgoal)
D9	P1 → AISA	*Performance Analytics* (resource)
D10–D15	R1–R3 ↔ P1	Role mandates and role outcomes (see spec)
D16–D18	Customer ↔ AISA / P1	Relational closure softgoals (see spec)

Relational closure. D16–D18 add direct Customer↔AISA and Customer↔P1 stakes alongside **D1–D15**.

Jake's boundary—Expert Contributor

- Jake's goal *Customer Issues be Resolved* (J-G1) uses means-ends **OR**: **J-T-ALT-A** *Resolve via Passive AI Compliance* vs. **J-T-ALT-B** *Resolve Using Expert Judgment with AI Support.* Under **Alternative B**, an AND decomposition from J-T-ALT-B yields J-T1 *Provide Real-Time Chat Support*, J-T2 *Handle Escalated Cases*, J-T3 *Mentor Through AI-Augmented Review*, J-T4 *Validate AI Suggestions.*
- **J-T4** refines into *Flag Incorrect Recommendations* and *Annotate Edge Cases with Rationale*—making Jake's expertise visible and reciprocal; typal **D5** names reciprocal resources *Validated Knowledge* and *Expert Annotations.*
- Softgoals: J-S1 *Professional Expertise be Recognized* (expert distinction and standing), J-S2 *Mentoring Authority be Sustained*, J-S3 *Knowledge Contribution be Attributed.*

	From	To	Label / rationale
Alternative A			
C-JA1	J-T-ALT-A	J-S1	Hurt: expertise not exercised
C-JA2	J-T-ALT-A	J-S2	Break: mentoring authority undermined
Alternative B			
C-J1	J-T4	J-S1	Make
C-J2	J-T2	J-S1	Help
C-J3	J-T3	J-S2	Help
C-J4	J-T4	J-S3	Make

Mia's boundary—Developing Professional

- M-G1 *Customer Issues be Resolved*: OR between **M-T-ALT-A** *Follow AI Templates Uncritically* and **M-T-ALT-B** *Engage Critically with AI Guidance.* Under Alt B, AND from M-T-ALT-B yields M-T1–M-T4 as in the specification. Mia's use of the AI instantiates typal **D6** *Transparent Augmentation*; the concrete SR dependency is *Scaffolded Learning Guidance* (designed to diminish over time), not the parent delegation **D4** alone.

	From	To	Label
C-MA1	M-T-ALT-A	M-S1	Hurt
C-MA2	M-T-ALT-A	M-S3	Break
C-MA3	M-T-ALT-A	M-S2	Help (durable learning, narrow)
C-M1	M-T2	M-S1	Help
C-M2	M-T3	M-S1	Help
C-M3	M-T1	M-S2	Help
C-M4	M-T4	M-S2	Help
C-M5	M-T3	M-S3	Help
C-M6	M-T4	M-S3	Help

AI Chat Assistant's boundary—Identity-Aware System

- **AI-G1** *Support Augmentation Targets be Met* (hard goal): sole means–ends parent **AI-T0** *Perform Identity-Aware Support Augmentation* AND-decomposes AI-T1 *Generate Explanatory Suggestions*, AI-T2 *Calibrate Support Level*, AI-T3 *Attribute Knowledge Sources.* **AI-T1** supplies reasoning alongside recommendations, satisfying the transparency requirement category (cf. CleverLoan) at the level of task design.

- Softgoals: AI-S1 *Expert Knowledge be Attributed*, AI-S2 *Novice Autonomy be Developed.*
- Internal contributions: C-AI1 AI-T3 → AI-S1; C-AI2 AI-T2 → AI-S2; C-AI3 AI-T1 → AI-S2.
- Typal instantiation: **D4**, reciprocal **D5**/**D6**, **D9**.

Support Operations (P1)—typal position

CS-2 does not draw an SR interior for P1. Obligations are **D7–D9**, **D10–D15**, **D17–D18** (registry in spec). Identity-aware governance (e.g., workforce commitments, role mandates, customer priorities in policy) is carried on these typal links rather than as a separate labeled SR subtree for the position.

Customer dependencies (external, typal layer)

D1 *Customer Issues be Resolved* and reciprocal **D2** *Customer Trust be Sustained* are preserved. Relational closure **D16–D18** states customer stakes in AI-mediated experience, governance voice, and policy transparency—substantiating differentiated service and trust beyond homogenized templates when combined with agent softgoals such as **M-S3** and role softgoals **R1-S1**.

Agent Mentoring (R3)—means-ends, tasks, and contributions

R3-G1 *Staff Development Targets be Met* (hard goal): OR between **R3-T-ALT-A** *Automate Guidance via AI* and **R3-T-ALT-B** *AI-Augmented Human Mentoring.* Under Alt B: R3-T1 *Review and Provide Expert Feedback*, R3-T2 *Model Diagnostic Reasoning.* Softgoals R3-S1 *Professional Socialization be Sustained*, R3-S2 *Mentoring Quality be Sustained*; org softgoal **D15** *Development Outcomes be Achieved.*

	From	To	Label
C-R3A1	R3-T-ALT-A	R3-S1	Break
C-R3A2	R3-T-ALT-A	J-S2 (Jake)	Break
C-R3A3	R3-T-ALT-A	R3-S2	Hurt
C-R3B1	R3-T1	R3-S1	Make
C-R3B2	R3-T1	R3-S2	Help
C-R3B3	R3-T2	R3-S1	Help
C-R3B4	R3-T-ALT-B	J-S2 (Jake)	Make

Cross-boundary contribution links (X1–X4)

	From	To	Label / role
X1	AI-T3	J-S3	Help
X2	AI-T2	M-S1	Help
X3	R3-T-ALT-A	J-S2	Break
X4	R3-T-ALT-B	J-S2	Make

Note: **X3**/**X4** link R3 means-ends alternatives to Jake's *Mentoring Authority be Sustained*—a distributed identity effect across boundaries. Qualitative label propagation is deferred; see CS-2 specification.

Dependency satisfaction (design closure)

Under Alternative B, typal dependencies **D1–D18** (including **D16–D18** relational closure) are intended to cohere with the SR tasks in the CS-2 specification. Full propagation analysis is out of scope for this document revision.

[CS-2 To-Be SR diagram]

Place `CS-2_ToBe_SR.pdf` beside this `.tex` file to compile the figure automatically. The SR layout is fully specified in `CS-2_ToBe_SR_specification.md` (constructs, D1–D18, alternatives).

Figure 2: Identity-sensitive To-Be SR model—customer support redesign.

Key analytical point. The To-Be SR demonstrates that addressing the identity concerns surfaced in CS-1 does not require sacrificing the productivity gains that motivated the AI deployment. The redesign preserves the AI's augmentative function—Mia still benefits from expert-derived suggestions, the organization still achieves throughput improvements—while restructuring five critical dimensions:

1. **Role portfolio restoration and formalization through INS → ISA → OCCUPIES → COVERS**—Jake's three-role portfolio (Supporting Customer + Escalation Handling + Agent Mentoring) is preserved, the previously informal roles are brought under organizational COVERS scope, and all direct PLAYS links are replaced by an inheritance chain so that every agent—human and AI—accesses roles through the organizational position. The ISA split (Human vs. AI Support Agent) unifies governance while preserving the type distinction that grounds identity analysis.

2. **Knowledge extraction becomes reciprocal**—Jake is an explicit dependee for validated knowledge, not a passive source of training data.

3. **AI dependency becomes scaffolded**—Mia's interaction with the AI is designed to cultivate autonomous judgment and eventual multi-role capability rather than permanent single-role reliance.

4. **Communication homogenization is counterbalanced**—agent softgoal *Communication Authenticity be Maintained* (**M-S3**) and typal **D16–D18** add customer-facing and governance stakes against convergence.

5. **Means-ends alternatives and contribution links**—OR decompositions with contrasting contribution profiles make explicit the identity cost of design choices that throughput metrics treat as equivalent; qualitative label propagation may be added in a later pass.

The four requirement categories from the paper's Table 2 are all exercised:

- **Transparency: AI-T1** *Generate Explanatory Suggestions*; typal **D6** *Transparent Augmentation.*
- **Authority preservation: J-T2**, **R2-T1**/**R2-T2**, **R2-G1**; **M-T2**, **M-S1**.
- **Professional growth: M-S2**, **R3-G1**, **D15**, **J-S1**/**J-S3**.
- **Expert recognition: J-S3**, **AI-S1**, **C-J4**, **X1**, typal **D5**, **D7**, **D8**.

What this reveals beyond the other cases. The CustomerSupport To-Be SR tests the framework's ability to model a prescriptive redesign for a case where the *original deployment was measured as a success* by conventional metrics. In Santander, identity-sensitive design was already present (the TA manager championed it). In CleverLoan, the failure was evident (destructive behaviors, data manipulation). In NeuroYou, identity tensions surfaced through visible professional conflict. The CustomerSupport case is the most analytically demanding because the identity concerns are *invisible to the organization*—the system works, productivity rises, customer sentiment improves, and yet professional identities are being reshaped in ways that conventional evaluation cannot detect. The To-Be SR demonstrates that identity-sensitive modeling can surface requirements that no stakeholder has articulated, in a system that no stakeholder perceives as broken.

5 Coverage Analysis: What CustomerSupport Adds

Following the software testing analogy, the table below tracks which aspects of the identity-sensitive modeling framework are exercised by each case.

Summary: CustomerSupport adds nine new test paths: (1) multi-role portfolio erosion—the AI compresses or displaces distinctive roles while augmenting the shared routine role, (2) AI as tacit-knowledge extractor without expert participation, (3) identity erosion through skill-level compression rather than task substitution, (4) role-level displacement of Agent Mentoring from human actor to AI, (5) prescriptive redesign of a deployment that conventional metrics judge as successful, (6) expert recognition and knowledge attribution as an identity requirement, (7) scaffolded progressive autonomy as an alternative to permanent AI dependency, (8) means-ends OR decompositions with contrasting contribution profiles, together with typal Strategic Dependencies D1–D18 (including D16–D18 relational closure on Customer, AI, and Support Operations), and (9) agent type hierarchy (INS→ISA→OCCUPIES→COVERS)—a structured inheritance chain that unifies human and AI agents under the same organizational governance while preserving the type distinction that grounds identity analysis. Together with the other three cases, it completes a four-trajectory typology: designed preservation (Santander), unaddressed destruction (CleverLoan), emergent transformation (NeuroYou), and **unintended homogenization with identity-sensitive remedy** (CustomerSupport).

6 Domain-Specific Differences from Other Cases

Modeling capability	Santander	Clever Loan	NeuroYou	Customer support	Notes
Person-based identity (agent)	Tested (Tim)	Tested (Max, Anna)	Tested (Leo, Diana)	Tested (Jake, Mia)	All cases
Divergent identities in same role	—	Tested (Max vs. Anna)	—	Tested (Jake vs. Mia)	CL + CS
AI as substitutive threat	—	Tested	—	—	CL only
AI as augmentative tool	Tested	—	—	Tested	S + CS
Multi-role portfolio erosion	—	—	—	**Tested (3 roles → 1 effective)**	**New**
AI as knowledge extractor/ redistributor	—	—	—	**Tested**	**New**
Identity erosion through skill homogenization	—	—	—	**Tested**	**New**
Invisible mentoring displacement	—	—	—	**Tested**	**New**
Prescriptive redesign of a "successful" system	—	—	—	**Tested**	**New**
Bilateral identity threat (cross-profession)	—	—	Tested	—	NY only
Productive + destructive in same actor	—	Tested	—	—	CL only
Transparency as met/unmet requirement	Met (S)	Unmet (CL)	—	**Met by redesign**	S + CS
Authority preservation	Met (S)	Unmet (CL)	—	**Met by redesign**	S + CS
Expert recognition/ attribution	—	—	—	**Tested**	**New**
Scaffolded progressive autonomy	—	—	—	**Tested**	**New**
Communication authenticity vs. homogenization	—	—	—	**Tested**	**New**
Means-ends alternatives + typal D1–D18 (relational closure)	—	—	—	**Tested**	**New**
Agent type hierarchy (INS→ISA→ OCCUPIES→ COVERS)	—	—	—	**Tested**	**New**
Emergent mutual learning	—	—	Tested	—	NY only
Identity trajectory	Preservation	Destruction	Transformation	**Homogenization → Redesign**	Unique

Table 1: Coverage analysis: CustomerSupport in the identity-sensitive modeling framework.

Dimension	Santander	Clever Loan	NeuroYou	Customer support
AI type	Augmentative (CV screening)	Substitutive (loan decisions)	Collaborative (ML hiring model)	Augmentative (chat suggestions)
Scale	Small team	Small unit	Two organizations	5,000+ agents
Source method	Qualitative	Qualitative	Ethnographic	Quantitative
Identity visibility	Surfaced by manager	Surfaced by resistance	Surfaced by conflict	**Invisible—surfaced only by modeling**
Organizational response	Designed mitigation	No mitigation	Negotiated adaptation	No identity-aware response
Our modeling role	Retrospective (success)	Diagnostic (failure)	Analytical (negotiation)	**Prescriptive (redesign)**
Temporal arc	4-year progression	Ongoing tension	Iterative mutual learning	Staggered rollout, ongoing